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(54) **SILICON CARBIDE JUNCTION BARRIER SCHOTTKY DIODE WITH ENHANCED RUGGEDNESS**

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CPC ***H10D 8/60*** (2025.01); ***H10D 8/051*** (2025.01); ***H10D 8/50*** (2025.01); ***H10D 62/102*** (2025.01); ***H10D 62/8325*** (2025.01)

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(Continued)

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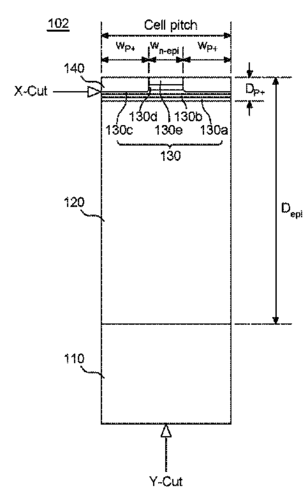
(57) **ABSTRACT**

Silicon carbide junction barrier Schottky diode disclosed. Silicon carbide junction barrier Schottky diode includes a first conductivity-type substrate, a first conductivity-type epitaxial layer, being formed by epitaxial growth of silicon carbide doped with a first conductivity-type impurity on the first conductivity-type substrate, a charge injection region, being formed on the first conductivity-type epitaxial layer and doped at a concentration of the first conductivity-type impurity higher than that of the first conductivity-type epitaxial layer, a second conductivity-type junction region, being formed on the first conductivity-type epitaxial layer so
(Continued)



as to contact the charge injection region, a Schottky metal layer, being formed on the charge injection region and the second conductivity-type junction region, an anode electrode, being formed on the Schottky metal layer, and a cathode electrode, being formed under the first conductivity-type substrate.

12 Claims, 11 Drawing Sheets

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See application file for complete search history.

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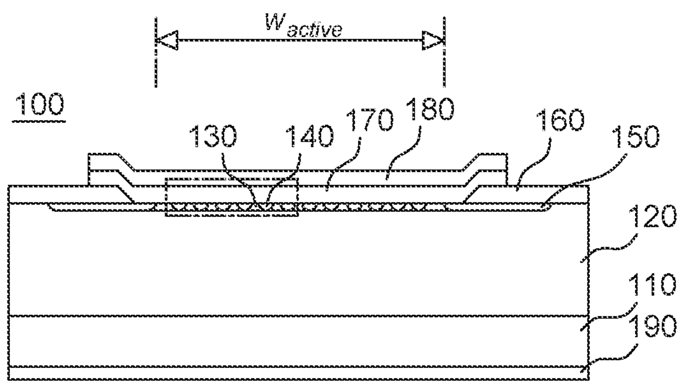


FIG. 1

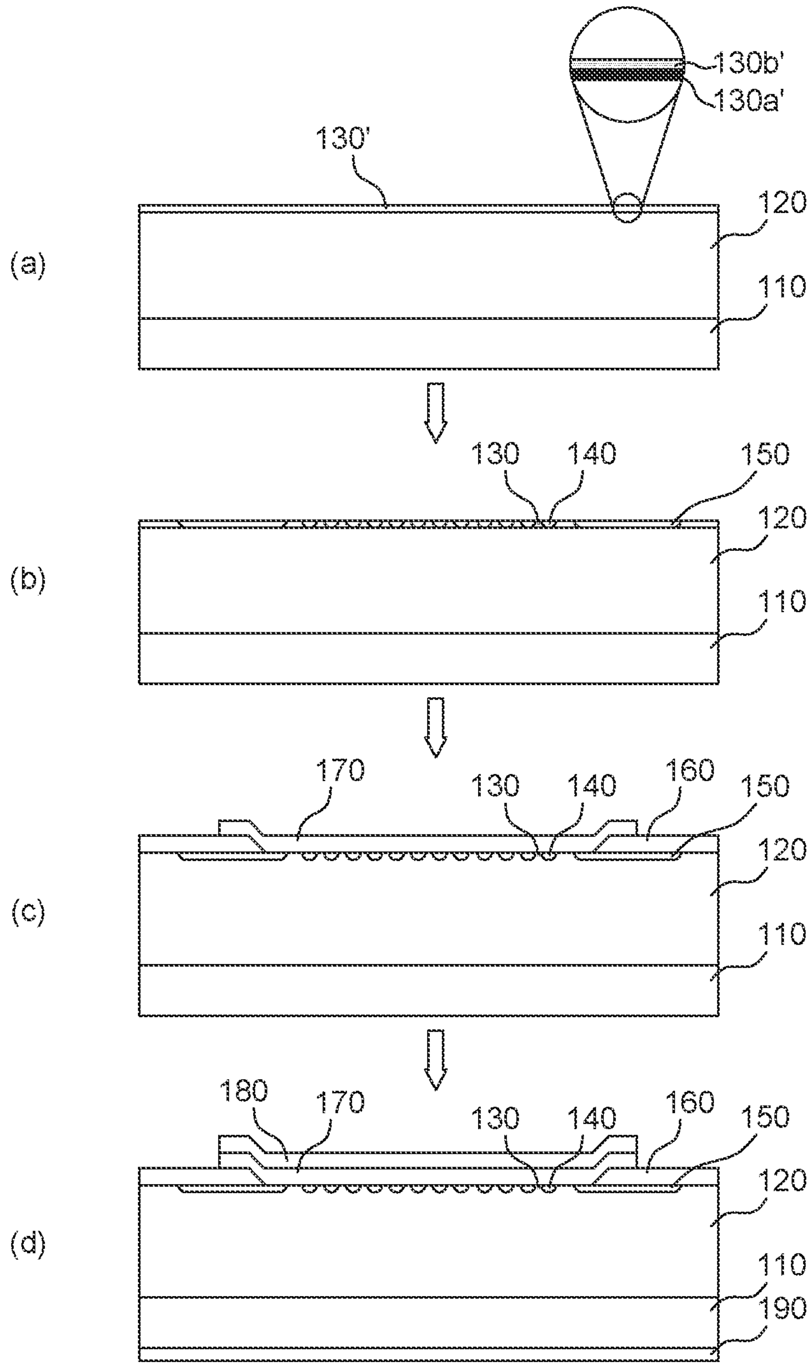

130b'
130a'
130'
120
(a)
110
130
140
150
120
(b)
110
170
130
140
160
150
120
(c)
110
180
170
130
140
160
150
120
(d)
110
190

FIG. 2

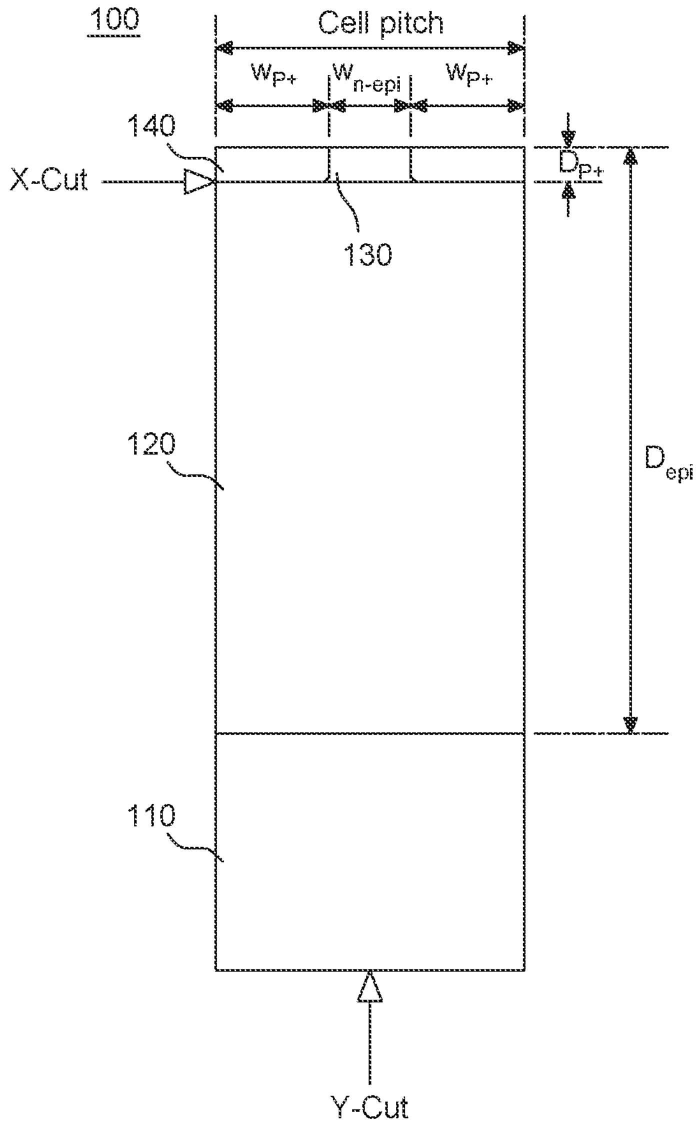

100
Cell pitch
W_{P+}
$W_{n\text{-}epi}$
W_{P+}
140
X-Cut
D_{P+}
130
120
D_{epi}
110
Y-Cut

FIG. 3

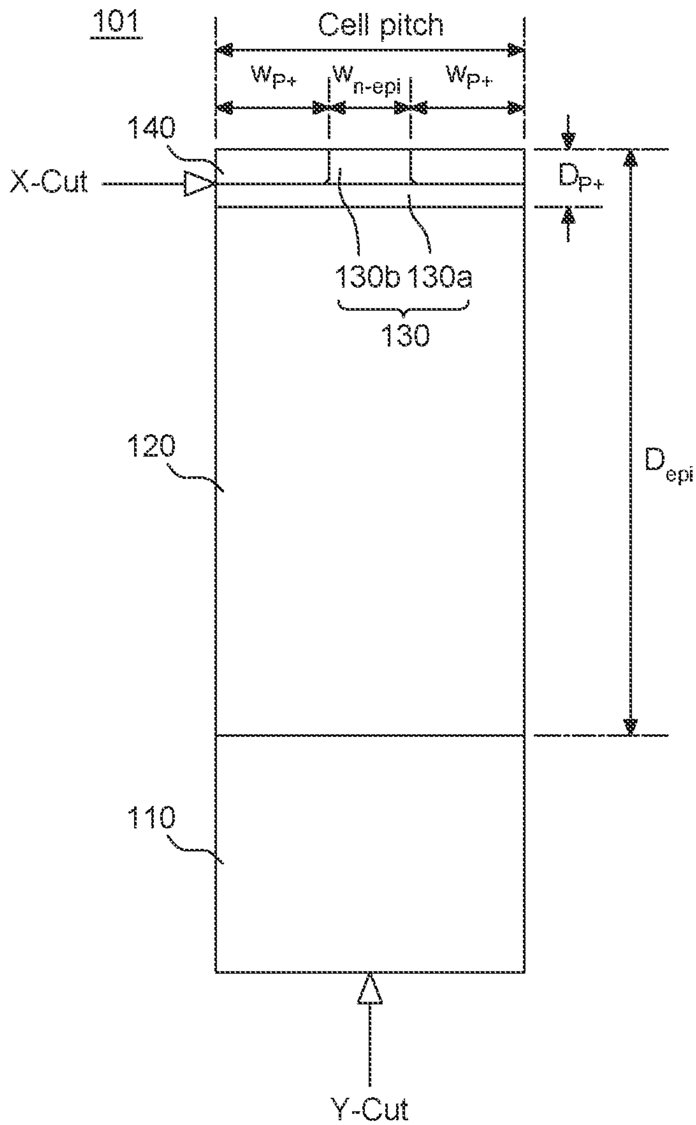

101
Cell pitch
W_P+
W_n-epi
W_P+
140
X-Cut
D_P+
130b
130a
130
120
D_epi
110
Y-Cut

FIG. 4

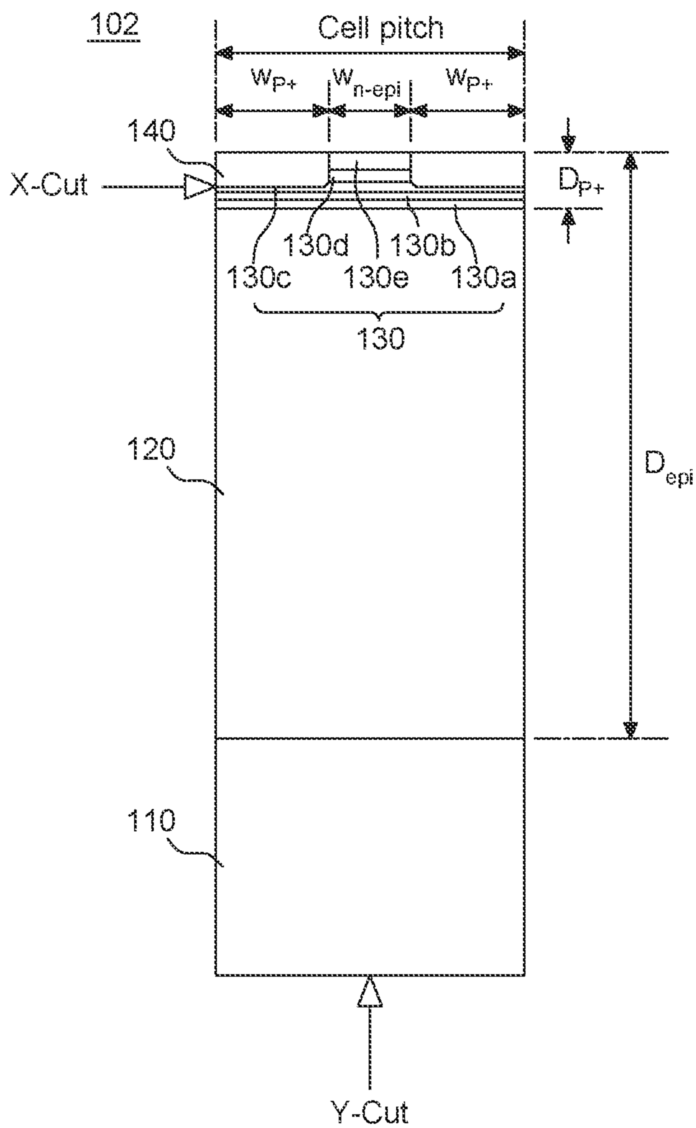

102
Cell pitch
W_{P+}
$W_{n\text{-}epi}$
W_{P+}
140
X-Cut
D_{P+}
130d
130b
130c
130e
130a
130
120
D_{epi}
110
Y-Cut

FIG. 5

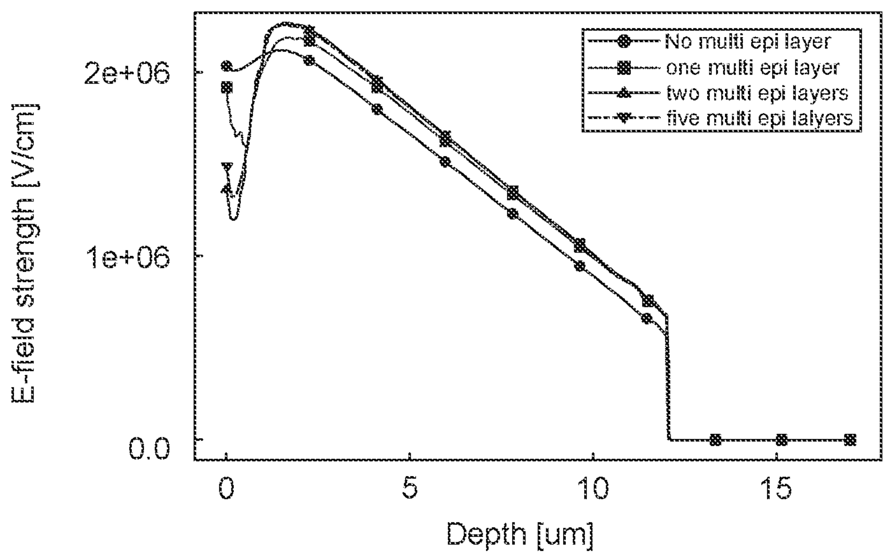

No multi epi layer
one multi epi layer
two multi epi layers
five multi epi lalyers
2e+06
1e+06
0.0
E-field strength [V/cm]
0
5
10
15
Depth [um]

FIG. 6

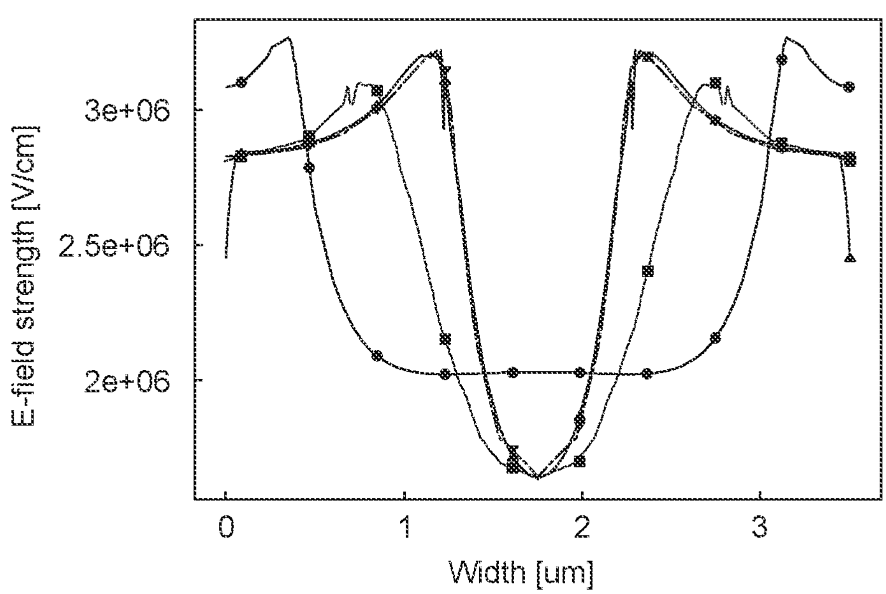

3e+06
2.5e+06
2e+06
E-field strength [V/cm]
0
1
2
3
Width [um]

FIG. 7

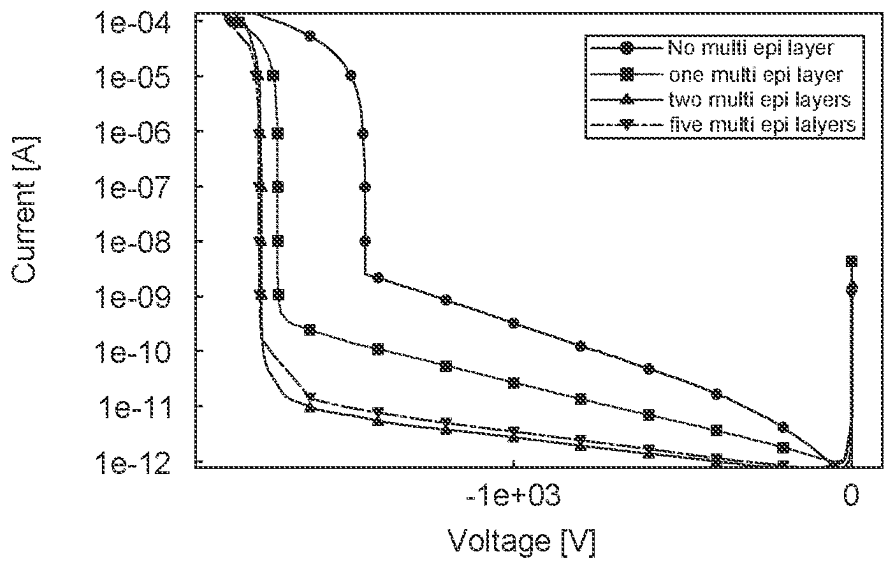

1e-04
1e-05
1e-06
1e-07
1e-08
1e-09
1e-10
1e-11
1e-12
Current [A]
-1e+03
0
Voltage [V]
No multi epi layer
one multi epi layer
two multi epi layers
five multi epi lalyers

FIG. 8

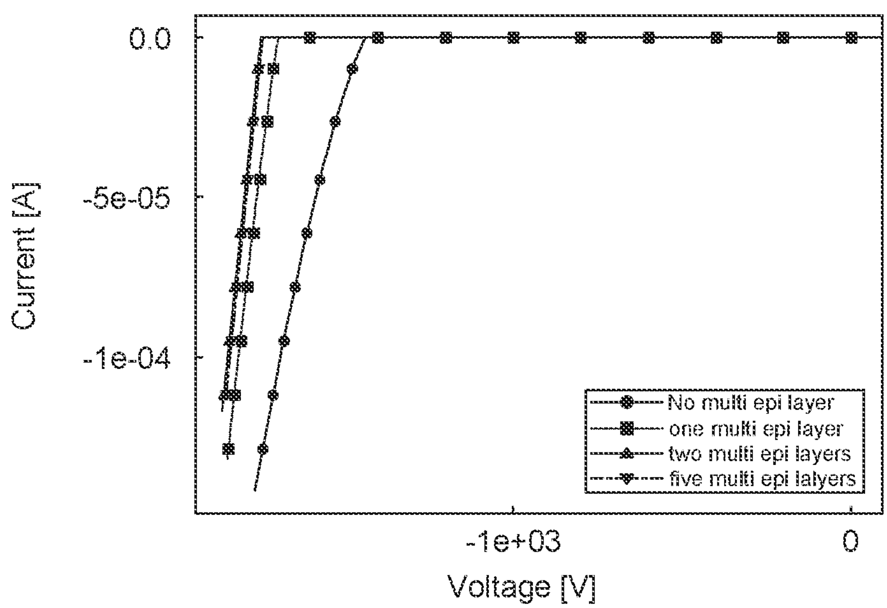

0.0
-5e-05
-1e-04
Current [A]
-1e+03
0
Voltage [V]
No multi epi layer
one multi epi layer
two multi epi layers
five multi epi lalyers

FIG. 9

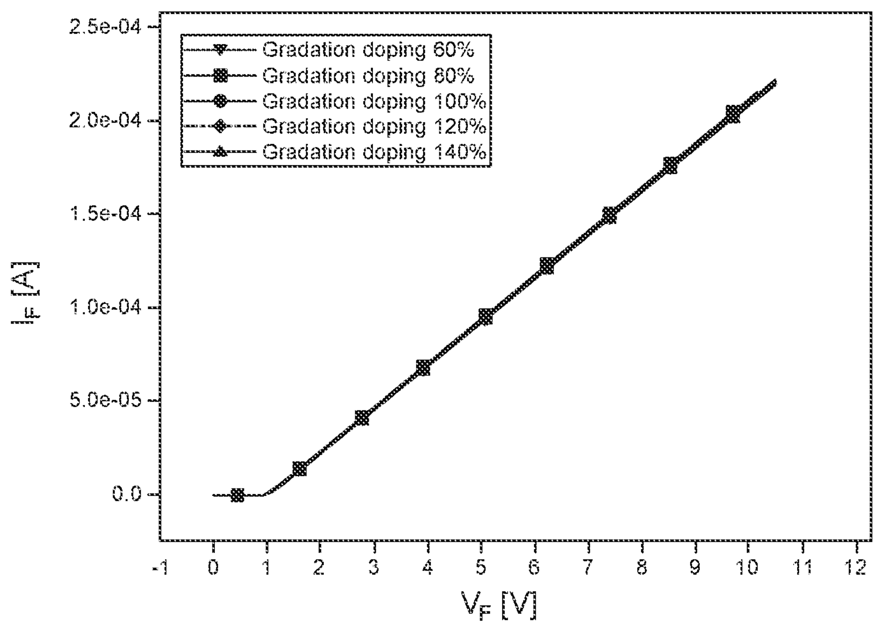

Gradation doping 60%
Gradation doping 80%
Gradation doping 100%
Gradation doping 120%
Gradation doping 140%
2.5e-04
2.0e-04
1.5e-04
1.0e-04
5.0e-05
0.0
I_F [A]
-1
0
1
2
3
4
5
6
7
8
9
10
11
12
V_F [V]

FIG. 10

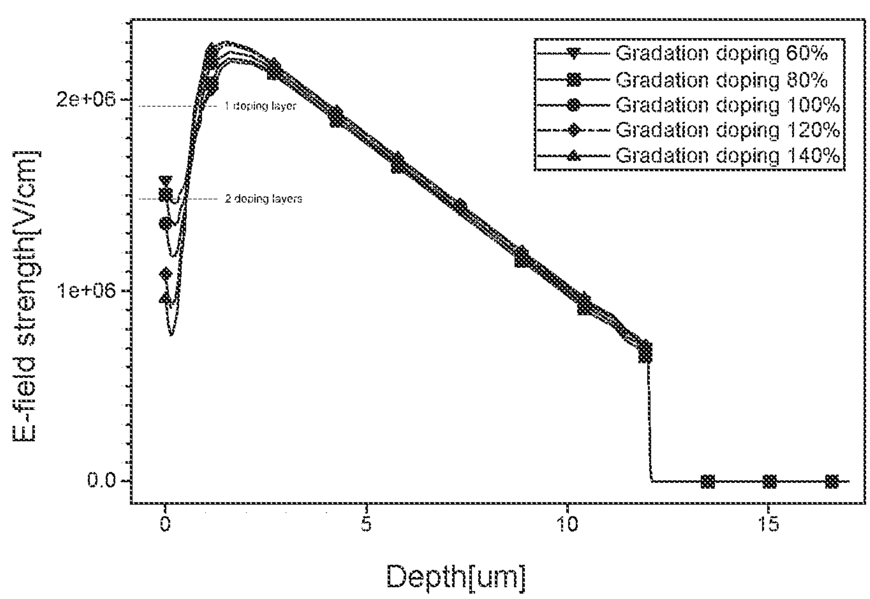
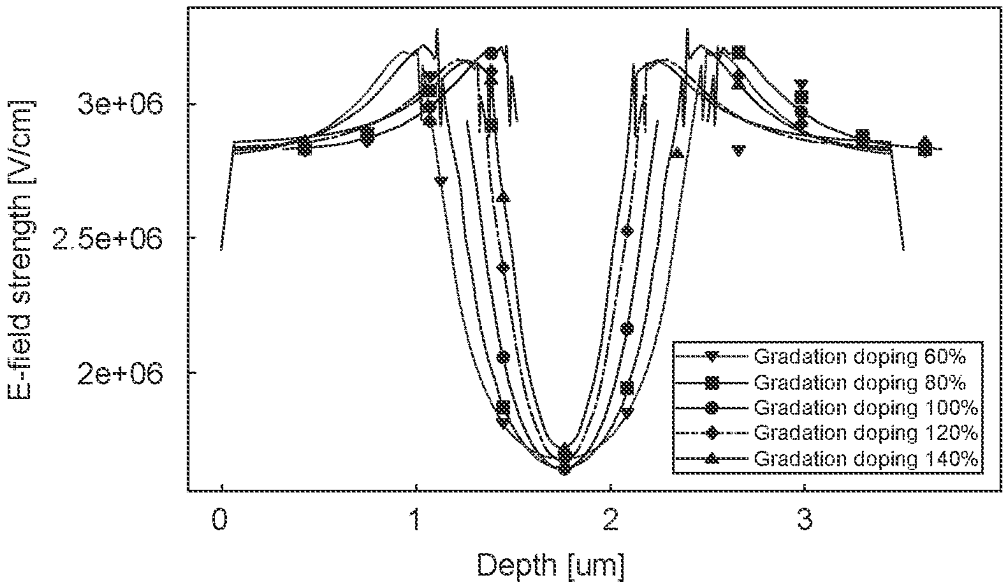
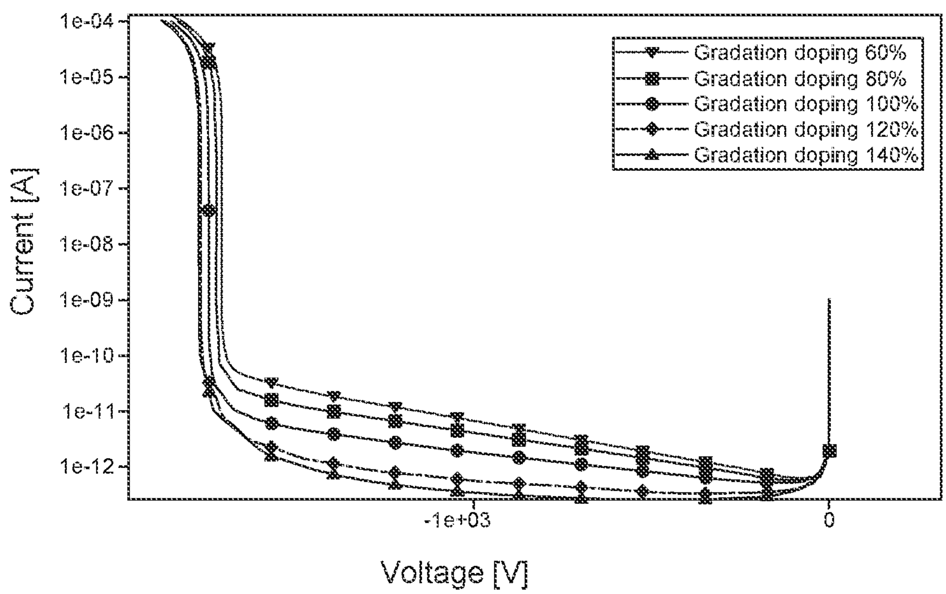

Gradation doping 60%
Gradation doping 80%
Gradation doping 100%
Gradation doping 120%
Gradation doping 140%
1 doping layer
2 doping layers
2e+06
1e+06
0.0
E-field strength[V/cm]
0
5
10
15
Depth[um]

FIG. 11



E-field strength [V/cm]
3e+06
2.5e+06
2e+06
Depth [um]
0
1
2
3
Gradation doping 60%
Gradation doping 80%
Gradation doping 100%
Gradation doping 120%
Gradation doping 140%

FIG. 12



Current [A]
1e-04
1e-05
1e-06
1e-07
1e-08
1e-09
1e-10
1e-11
1e-12
Voltage [V]
-1e+03
0
Gradation doping 60%
Gradation doping 80%
Gradation doping 100%
Gradation doping 120%
Gradation doping 140%

FIG. 13

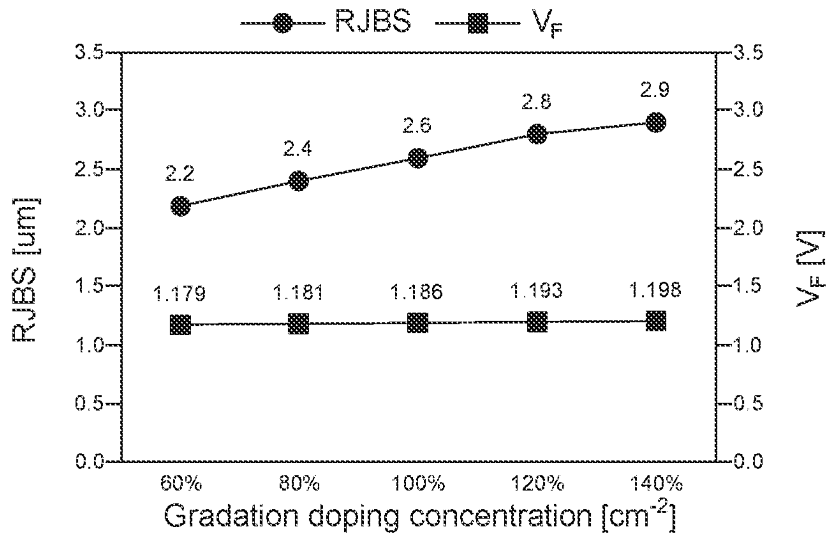

RJBS
V_F
RJBS [um]
V_F [V]
2.2
2.4
2.6
2.8
2.9
1.179
1.181
1.186
1.193
1.198
60%
80%
100%
120%
140%
Gradation doping concentration [cm^{-2}]

FIG. 14

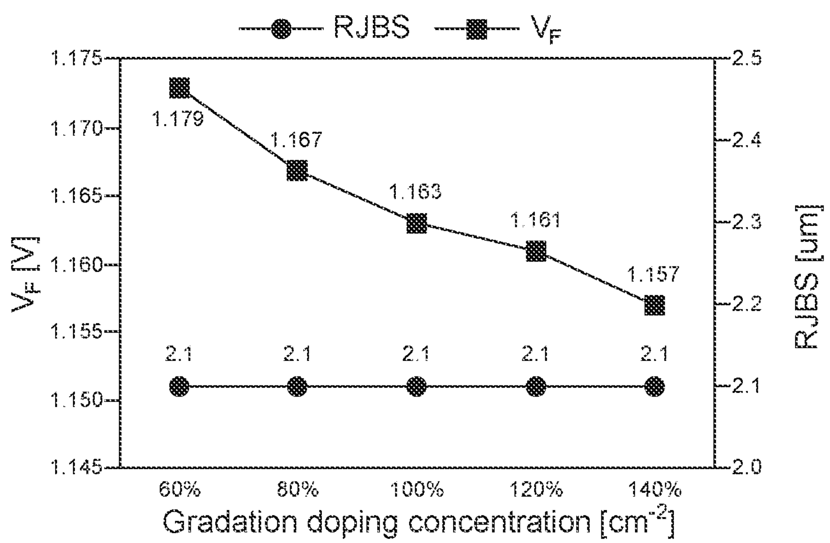

RJBS
V_F
V_F [V]
RJBS [um]
1.179
1.167
1.163
1.161
1.157
2.1
2.1
2.1
2.1
2.1
60%
80%
100%
120%
140%
Gradation doping concentration [cm^{-2}]

FIG. 15

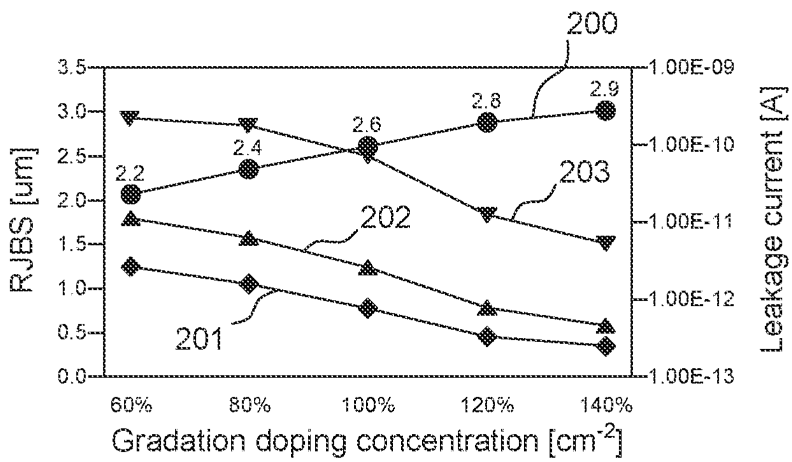

200
203
202
201
2.2
2.4
2.6
2.8
2.9
RJBS [um]
Leakage current [A]
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
1.00E-09
1.00E-10
1.00E-11
1.00E-12
1.00E-13
60%
80%
100%
120%
140%
Gradation doping concentration [cm-2]

FIG. 16

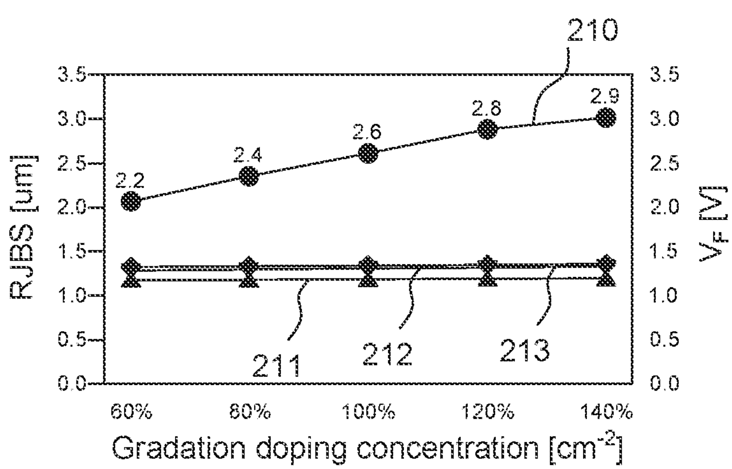

210
211
212
213
2.2
2.4
2.6
2.8
2.9
RJBS [um]
V_F [V]
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
60%
80%
100%
120%
140%
Gradation doping concentration [cm-2]

FIG. 17

SILICON CARBIDE JUNCTION BARRIER SCHOTTKY DIODE WITH ENHANCED RUGGEDNESS

FIELD

Present invention relates to a power semiconductor.

BACKGROUND

Silicon carbide junction barrier Schottky diode (SiC SBD) has characteristics of low turn-on voltage, junction capacitance, fast recovery time, and high cutoff frequency. Due to these characteristics, SiC SBD is widely used in radio frequency switched-mode power supply (RF SMPS), power management integrated circuit (PIMC), and the like. In order to improve the maximum forward surge current IFSM characteristics and electro-static discharge (ESD) characteristics of the SiC SBD, the width of the P+ region shall be significant. However, if the P+ region is significantly wide, the width of the SBD region is relatively reduced, thereby reducing the forward current.

SUMMARY

According to one aspect of the embodiment according to the present invention, a unit cell constituting an active area of silicon carbide junction barrier Schottky diode is provided. The unit cell includes a first conductivity-type substrate, a first conductivity-type epitaxial layer, being formed by epitaxial growth of silicon carbide doped with a first conductivity-type impurity on the first conductivity-type substrate, a charge injection region, being defined by a second conductivity-type junction region that is formed by ion implantation of a second conductivity-type impurity into a first conductivity-type charge injection layer that is formed by epitaxial growth of silicon carbide doped at a concentration of the first conductivity-type impurity higher than that of the first conductivity-type epitaxial layer on the first conductivity-type epitaxial layer, a Schottky metal layer, being formed on the charge injection region and the second conductivity-type junction region, an anode electrode, being formed on the Schottky metal layer, and a cathode electrode, being formed under the first conductivity-type substrate.

In one embodiment, an area occupied by the second conductivity-type junction region in the unit cell is in a range between 62% and 82%.

In one embodiment, the first conductivity-type charge injection layer includes two or more multi-epitaxial layers having different first conductivity-type impurity concentrations.

In one embodiment, the first conductivity-type impurity concentrations of the two or more multi-epitaxial layers according to a reference concentration profile decrease in a direction toward the first conductivity-type epitaxial layer, wherein the reference concentration profile defines the first conductivity-type impurity concentrations of the two or more multi-epitaxial layers.

In one embodiment, the reference concentration profile decreases the impurity concentration by 20% in the direction toward the first conductivity-type epitaxial layer.

In one embodiment, the first conductivity-type impurity concentrations of each of the two or more multi-epitaxial layers are equally increased by a maximum of 40% or a decrease of a minimum of −40% in the reference concentration profile.

In one embodiment, the depths of two or more multi-epitaxial layers decrease in the direction toward the first conductivity-type epitaxial layer.

In one embodiment, a depth of the second conductivity-type junction region is in a range between 60% and 100% of a depth of the charge injection region.

According to another aspect of the embodiment according to the present invention, a silicon carbide junction barrier Schottky diode having an active area is provided. The silicon carbide junction barrier Schottky diode includes a first conductivity-type substrate, a first conductivity-type epitaxial layer, being formed by epitaxial growth of silicon carbide doped with a first conductivity-type impurity on the first conductivity-type substrate, a charge injection region, being formed on the first conductivity-type epitaxial layer and doped at a concentration of the first conductivity-type impurity higher than that of the first conductivity-type epitaxial layer, a second conductivity-type junction region, being formed on the first conductivity-type epitaxial layer so as to contact the charge injection region, a Schottky metal layer, being formed on the charge injection region and the second conductivity-type junction region, an anode electrode, being formed on the Schottky metal layer, and a cathode electrode, being formed under the first conductivity-type substrate.

BRIEF DESCRIPTION OF THE DRAWINGS

Hereinafter, embodiments of the invention will be described with reference to the accompanying drawings. For the purpose of easy understanding of the invention, the same elements will be referred to by the same reference signs. Configurations illustrated in the drawings are examples for describing the invention, and do not restrict the scope of the invention. Particularly, in the drawings, some elements are slightly exaggerated for the purpose of easy understanding of the invention. Since the drawings are used to easily understand the invention, it should be noted that widths, depthes, and the like of elements illustrated in the drawings might change at the time of actual implementation thereof. Meanwhile, throughout the detailed description of the invention, the same components are described with reference to the same reference numerals.

FIG. **1** is a schematic cross-sectional view of SiC SBD with enhanced ruggedness;

FIG. **2** exemplarily illustrates a process of manufacturing the SiC SBD shown in FIG. **1**;

FIG. **3** is a cross-sectional view exemplarily illustrating one embodiment of a unit cell constituting the active area of the SiC SBD shown in FIG. **1**;

FIG. **4** is a cross-sectional view exemplarily illustrating another embodiment of a unit cell constituting the active area of the SiC SBD shown in FIG. **1**;

FIG. **5** is a cross-sectional view exemplarily illustrating still another embodiment of a unit cell constituting the active area of the SiC SBD shown in FIG. **1**;

FIG. **6**, FIG. **7**, FIG. **8** and FIG. **9** are graphs exemplarily illustrating electrical characteristics of the SiC SBD shown in FIGS. **3** to **5**;

FIG. **10**, FIG. **11**, FIG. **12** and FIG. **13** are graphs exemplarily illustrating electrical characteristics according to a change in doping concentration of the SiC SBD shown in FIG. **5**;

FIG. **14** and FIG. **15** are graphs exemplarily illustrating a function of the charge injection region; and

FIG. **16** and FIG. **17** are graphs exemplarily illustrating electrical characteristics according to the depth of the second conductivity-type junction region.

DETAILED DESCRIPTION

Embodiments which will be described below with reference to the accompanying drawings can be implemented singly or in combination with other embodiments. But this is not intended to limit the present invention to a certain embodiment, and it should be understood that all changes, modifications, equivalents or replacements within the spirits and scope of the present invention are included. Especially, any of functions, features, and/or embodiments can be implemented independently or jointly with other embodiments. Accordingly, it should be noted that the scope of the invention is not limited to the embodiments illustrated in the accompanying drawings.

Terms such as first, second, etc., may be used to refer to various elements, but these element should not be limited due to these terms. These terms will be used to distinguish one element from another element.

The terms used in the following description are intended to merely describe specific embodiments, but not intended to limit the invention. An expression of the singular number includes an expression of the plural number, so long as it is clearly read differently. The terms such as "include" and "have" are intended to indicate that features, numbers, steps, operations, elements, components, or combinations thereof used in the following description exist and it should thus be understood that the possibility of existence or addition of one or more other different features, numbers, steps, operations, elements, components, or combinations thereof is not excluded.

When an element, such as a layer, is referred to as being "on," "connected to," or "coupled to" another element or layer, it may be directly on, connected to, or coupled to the other element or layer or intervening elements or layers may be present. When, however, an element or layer is referred to as being "directly on," "directly connected to," or "directly coupled to" another element or layer, there are no intervening elements or layers present.

Spatially relative terms, such as "beneath," "below," "lower," "above," "upper," and the like, may be used herein for descriptive purposes, and, thereby, to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the drawings. Spatially relative terms are intended to encompass different orientations of an apparatus in use, operation, and/or manufacture in addition to the orientation depicted in the drawings.

FIG. **1** is a schematic cross-sectional view of silicon carbide junction barrier Schottky diode (SiC SBD) with enhanced ruggedness.

SiC SBD **100** includes an active area and an edge termination area surrounding at least a portion of the active area. The active area includes a Schottky junction and a PN junction. The edge termination area may have various structures such as a guard ring, a bevel, and a junction termination extension.

SiC SBD **100** includes a first conductivity-type substrate **110**, a first conductivity-type epitaxial layer **120**, a charge injection region **130**, a second conductivity-type junction region **140**, a second conductivity-type buffer region **150**, an insulating layer **160**, a Schottky metal layer **170**, an anode electrode **180**, and a cathode electrode **190**. Here, the first conductivity-type refers to a region or a layer doped by an n-type impurity, and the second conductivity-type refers to the region or layer doped by a p-type impurity, but it can be understood that the reverse case is also possible.

The first conductivity-type substrate **110** may be formed by doping the first conductivity-type impurity on a **4**H-SiC or **6**H-SiC substrate. For example, the concentration of the first conductivity-type impurity in the first conductivity-type substrate **110** may be about 1E18 cm^{-2}, and the depth of the first conductivity-type substrate **110** may be about 5 μm.

The first conductivity-type epitaxial layer **120** may be formed by epitaxial growth of silicon carbide doped with the first conductivity-type impurity on the first conductivity-type substrate **110**. For example, the concentration of the first conductivity-type impurity in the first conductivity-type epitaxial layer **120** may be about 1E16 cm^{-2}, and the depth of the first conductivity-type epitaxial layer **120** may be about 11 μm.

The charge injection region **130** may be formed in the active area on an upper surface of the first conductivity-type epitaxial layer **120**. The charge injection region **130** may have a higher impurity concentration than the first conductivity-type epitaxial layer **120**. For example, the concentration of the first conductivity-type impurity in the charge injection region **130** may be at least 1.2E16 cm^{-2} up to 1.4E17 cm^{-2}, which may be at least 1.2 to 1.4 times the impurity concentration of the first conductivity-type epitaxial layer **120**. The depth of the charge injection region **130** may be about 0.6 μm to about 1.0 μm. Meanwhile, the impurity concentration of the charge injection region **130** may be different depending on the depth from the upper surface. For example, the impurity concentration of the charge injection region **130** may decrease as the distance from the upper surface increases.

As the concentration of the first conductivity-type impurity increases, a resistivity of the charge injection region **130** decreases. When the resistivity of the charge injection region **130** in contact with the Schottky metal layer decreases, an amount of charges passing through the Schottky junction increases and the current increases accordingly.

The plurality of second conductivity-type junction regions **140** may be formed in the active area on the upper surface of the first conductivity-type epitaxial layer **120**. The plurality of second conductivity-type junction regions **140** may be formed by an ion implantation of the second conductivity-type impurity into the first conductivity-type charge injection layer **130'** (see FIG. **2**) at a relatively high concentration. For example, the concentration of the second conductivity-type impurity implanted into the second conductivity-type junction region **140** may be about 1E18 cm^{-2}, and the depth of the second conductivity-type junction region **140** may be about 0.6 μm. By the ion implantation, the plurality of second conductivity-type junction regions **140** may be formed to extend from the upper surface of the first conductivity-type charge injection layer **130'** into the first conductivity-type charge injection layer **130'**. A ratio of area occupied by the second conductivity-type junction region **140** in the active area may be at least 62% and up to 82%.

The second conductivity-type buffer region **150** may be formed throughout the active area and the edge termination area on the upper surface of the first conductivity-type epitaxial layer **120** and surrounds the active area. Specifically, in a horizontal direction, one end of the second conductivity-type buffer region **150** extends beyond the insulating layer **160** and contacts the Schottky metal layer **170**, and the other end of the second conductivity-type buffer region **150** extends toward the edge termination area. The second conductivity-type buffer region **150** extending to the edge termination area contacts the insulating layer **160** but does not extend beyond the insulating layer **160**. The second conductivity-type buffer region **150** may be formed by ion implantation of the second conductivity-type impurity into the first conductivity-type charge injection layer **130'** (see FIG. **2**). The width of the second conductivity-type buffer region **150** may be wider than that of the second conductivity-type junction region **140**. The second conductivity-type buffer region **150** has a function of dispersing an electric field while maintaining a breakdown voltage of the device.

The insulating layer **160** may be formed on a portion of the buffer layer **150** and on the first conductivity-type epitaxial layer **120** in the edge termination area. The insulating layer **160** may be formed to surround the plurality of second conductivity-type junction regions **140** to define an active area. The insulating layer **160** may be formed to overlap at least a portion of the second conductivity-type buffer region **150**. The insulating layer **160** may be formed of silicon oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), borophosphosilicate glass (BPSG), and the like.

The Schottky metal layer **170** may be formed on the first conductivity-type epitaxial layer **120**, contacts the upper surfaces of the plurality of second conductivity-type junction regions **140**, and has Schottky junction with a portion of first conductivity-type epitaxial layer **120**. The Schottky metal layer **170** may extend in the horizontal direction to cover a portion of the insulating layer **160**.

The anode electrode **180** may be formed over the Schottky metal layer **170**, and the cathode electrode **190** may be formed under the first conductivity-type substrate **110**. The anode electrode **180** and the cathode electrode **190** may be formed of a metal or a metal alloy. A silicide layer (not shown) for ohmic contact may be formed between the first conductivity-type substrate **110** and the cathode electrode **190**.

The operation of the above-described SiC SBD **100** will be described.

The plurality of second conductivity-type junction regions **140** are PN-junctioned with the first conductivity-type charge injection region **130**. Accordingly, the plurality of second conductivity-type junction regions **140** and the first conductivity-type charge injection region **130** perform the same function as a PiN diode. That is, SiC SBD **100** has a structure in which a junction barrier Schottky diode (hereinafter referred to as SBD) and a PiN diode are connected in parallel.

In off state of SiC SBD **100**, the SBD and the PiN diode simultaneously maintain the off-state voltage. Since a depth from the lower portion of the PiN diode to the first conductivity-type substrate **110** is smaller than a depth from the lower portion of the SBD to the first conductivity-type substrate **110**, a relatively strong off-state electric field is applied to the PiN diode and, as a result, a relatively weak off-state electric field is applied to the SBD. In other words, due to the PiN diode, the electric field applied to an interface of the Schottky junction decreases, thereby reducing the leakage current.

The turn-on voltage of the PiN diode is relatively higher than that of the SBD. In on state, when a forward voltage V_F is applied to the PiN diode and SBD connected in parallel, current starts to flow through the SBD having a relatively low turn-on voltage. Then, when the forward voltage V_F increases above the turn-on voltage of the PiN diode, current flows through both the PiN diode and the SBD. For this reason, in the known JBS diode, the current density characteristic in the on state decreases compared to a conventional SBD in which current flows through the entire active area. On the other hand, in SiC SBD **100** according to one embodiment of the present invention, the charge injection region **130** increases the current density, and thus substantially the same current density as that of the known JBS diodes can be achieved, even if the area of the second conductivity-type junction region **140** increases.

FIG. **2** exemplarily illustrates a process of manufacturing the SiC SBD shown in FIG. **1**. The charge injection region **130** may be formed by the ion implantation or a multi-epitaxial process. Hereinafter, a process of manufacturing the SiC SBD by use of the multi-epitaxial process will be described.

In step (a), the first conductivity-type charge injection layer **130'** may be formed on the upper surface of the first conductivity-type epitaxial layer **120**. The first conductivity-type epitaxial layer **120** is epitaxially grown on the upper surface of the first conductivity-type substrate **110**. The first conductivity-type substrate **110** has a depth of about 5 μm, and may be doped with the first conductivity-type impurity at a concentration of about 1E18 cm^{-2}. Meanwhile, the depth of the first conductivity-type epitaxial layer **120** is about 11 μm and may be doped with the first conductivity-type impurity at a concentration of about 1E16 cm^{-2}.

The doping concentration of the first conductivity-type charge injection layer **130'** may be consistent regardless of depth or vary with depth. The first conductivity-type charge injection layer **130'** may be multi-epitaxial layers **130*a*'** and **130*b*'**. Doping concentrations of the first multi-epitaxial layer **130*a*'** and the second multi-epitaxial layer **130*b*'** may be different. For example, the doping concentration of the first multi-epitaxial layer **130*a*'** may be lower than that of the second multi-epitaxial layer **130*b*'**.

The depth of the first conductivity-type charge injection layer **130'** may be changed according to the number of multi-epitaxial layers **130*a*'** and **130*b*'**. The number of the multi epitaxial layers **130*a*'** and **130*b*'** may be 2 to 5, for example. Accordingly, the depth of the first conductivity-type charge injection layer **130'** may be about 0.6 μm to about 1.0 μm.

In step (b), a plurality of second conductivity-type junction regions **140** may be formed in the first conductivity-type charge injection layer **130'**. Meanwhile, the second conductivity-type buffer region **150** may be formed simultaneously with the plurality of second conductivity-type junction regions **140**. A photoresist is applied to the upper surface of the first conductivity-type epitaxial layer **120**, and the photoresist located in the region to be ion implanted is removed. Thereafter, the second conductivity-type impurity may be implanted. Meanwhile, a plurality of guard rings may be formed in the second conductivity-type buffer region **150**.

In step (c), the insulating layer **160** may be formed on the upper surface of the first conductivity-type epitaxial layer **120**, and the Schottky metal layer **170** may be formed in the active area. The insulating layer **160** may be formed by forming an insulating film of silicon oxide, PSG, BSG, BPSG, or the like on the entire upper surface of the first conductivity-type epitaxial layer **120**, and then removing the insulating film corresponding to the active area. Schottky metal layer **170** may be formed of a material having a low energy barrier or an alloy of two or more metals, for example, any one or a combination of titanium, chromium, polysilicon, aluminum, and tantalum.

In step (d), the anode electrode **180** may be formed on the Schottky metal layer **170** and the cathode electrode **190** may be formed under the first conductivity-type substrate **110**, respectively. The electrodes may be formed of, for example, aluminum.

FIG. **3** is a cross-sectional view exemplarily illustrating one embodiment of a unit cell constituting the active area of the SiC SBD shown in FIG. **1**.

Referring to FIG. **3**, the SiC SBD **100** may include the charge injection region **130** doped with a uniform concentration. The charge injection region **130** may be formed by doping, for example, the first conductivity-type impurity at a concentration of about 1E16 cm^{-2}. The cell pitch of the SiC SBD **100** is about 3.5 μm, and the ratio within the cell pitch between a width $W_{n\text{-}epi}$ of the charge injection region **130** and a width $2\times W_{P+}$ of the second conductivity-type junction region **140** is 1.0:2.5. Hereinafter, it is assumed that the cell pitch of the SiC SBD **100** is 3.5 μm, and the width of the second conductivity-type junction region **140** is referred to as RJBS. The width $W_{n\text{-}epi}$ of the charge injection region **130** may be about 1.0 μm and a depth D_{P+} may be about 0.6 μm. The RJBS may be about 2.5 μm, and a depth of the second conductivity-type junction region **140** may be about 0.6 μm. The second conductivity-type junction region **140** may be formed by doping, for example, the second conductivity-type impurity at a concentration of about 1E18 cm^2. Meanwhile, a depth D_{epi} from the upper surface of the first conductivity-type substrate **110** to the upper surface of the second conductivity-type junction region **140** may be about 12 μm.

The design parameters described above were selected for the purpose of comparison with the known JBS diode. Design parameters other than the depth of the charge injection region **130**, the first conductivity-type epitaxial layer **120**, and the RJBS are the same both in the known JBS diode and the SiC SBD **100**. In the known JBS diode, when the depth of the first conductivity-type epitaxial layer is about 12 μm, the RJBS is about 1.24 μm, and the forward voltage V_F is about 0.76 V, a current of about 10 A may flow. In order to flow the substantially same forward current by applying the substantially same forward voltage V_F, the RJBS of the SiC SBD **100** is selected to be about 2.5 μm.

FIG. **4** is a cross-sectional view exemplarily illustrating another embodiment of a unit cell constituting the active area of the SiC SBD shown in FIG. **1**.

Referring to FIG. **4**, the SiC SBD **101** may include the charge injection region **130** composed of two multi-epitaxial layers **130***a* and **130***b* doped with different concentrations. The first multi-epithelial layer **130***a* may be formed by doping the first conductivity-type impurity at a concentration of about 3E16 cm^{-2} and may have a depth of about 0.6 μm. The second multi epitaxial layer **130***b* may be formed by doping the first conductivity-type impurity at a concentration of about 1E17 cm^{-2} and may have a depth of about 0.4 μm. The cell pitch of the SiC SBD **101** may be about 3.5 μm, and the RJBS may be about 2.6 μm. The second conductivity-type junction region **140** may be formed by doping, for example, the second conductivity-type impurity at a concentration of about 1E18 cm^{-2}. The depth D_{P+} of the second conductivity-type junction region **140** may be about 0.6 μm. Meanwhile, the depth D_{epi} from the upper surface of the first conductivity-type substrate **110** to the upper surface of the second conductivity-type junction region **140** may be about 12 μm.

The design parameters described above were selected for the purpose of comparison with the known JBS diode. The design parameters other than the charge injection region **130** and RJBS are the same both in the known JBS diode and the SiC SBD **101**. In the known JBS diode, when the depth of the first conductivity-type epitaxial layer is about 12 μm, the RJBS is about 0.9 μm, and the forward voltage V_F is about 0.66 V, a current of about 10 A may flow. In order to flow a forward current of 10 A by applying a forward voltage V_F of 0.68 V, the RJBS of the SiC SBD **100** is selected to be about 2.6 μm.

FIG. **5** is a cross-sectional view exemplarily illustrating still another embodiment of a unit cell constituting the active area of the SiC SBD shown in FIG. **1**.

Referring to FIG. **5**, the SiC SBD **102** may have a charge injection region **130** composed of five multi-epitaxial layers **130***a*, **130***b*, **130***c*, **130***d*, and **130***e* doped with different concentrations. The first multi epitaxial layer **130***a* may be formed by doping the first conductivity-type impurity at a concentration of about 2E16 cm^{-2} and may have a depth of about 0.1 μm. The second multi epitaxial layer **130***b* may be formed by doping the first conductivity-type impurity at a concentration of about 4E16 cm^{-2} and may have a depth of about 0.15 μm. The third multi epitaxial layer **130***c* may be formed by doping the first conductivity-type impurity at a concentration of about 6E16 cm^{-2} and may have a depth of about 0.2 μm. The fourth multi epitaxial layer **130***d* may be formed by doping the first conductivity-type impurity at a concentration of about 8E16 cm^{-2} and may have a depth of about 0.2 μm. The fifth multi-epithelial layer **130***e* may be formed by doping the first conductivity-type impurity at a concentration of about 1E17 cm^{-2} and may have a depth of about 0.3 μm. The cell pitch of the SiC SBD **102** may be about 3.5 μm, and the RJBS may be about 2.6 μm. The second conductivity-type junction region **140** may be formed by doping, for example, the second conductivity-type impurity at a concentration of about 1E18 cm^{-2}. The depth D_{P+} of the second conductivity-type junction region **140** may be about 0.6 μm. Meanwhile, a depth from the lower surface of the charge injection region **130** to the upper surface of the first conductivity-type substrate **110**, that is, the depth of the first conductivity-type epitaxial layer **120** may be about 11 μm. The depth of the first conductivity-type epitaxial layer **120** may vary according to a desired breakdown voltage requirement. In general, the depth of the first conductivity-type epitaxial layer **120** in the case of 650 V class diode is designed to be about 5 to about 7 μm, the depth of the first conductivity-type epitaxial layer **120** in the case of 1200 V class diode is designed to be about 10 to about 12 μm, and the depth of the first conductivity-type epitaxial layer **120** in the case of 1700 V class diode is designed to be about 15 μm or more.

The design parameters described above were selected for the purpose of comparison with the known JBS diode. The design parameters other than the charge injection region **130** and RJBS are the same both in the known JBS diode and SiC SBD **102**. In the known JBS diode, when the depth of the first conductivity-type epitaxial layer is about 12 μm, the RJBS is about 0.9 μm, and the forward voltage V_F is about 1.16 V, a current of about 10 A may flow. In order to flow a forward current of 10 A when the forward voltage V_F is about 1.18 V, the RJBS of the SiC SBD **102** is selected to be about 2.6 μm.

FIG. **6**, FIG. **7**, FIG. **8** and FIG. **9** are graphs exemplarily illustrating electrical characteristics of the SiC SBD shown in FIGS. **3** through **5**.

FIG. **6** shows vertical electric fields applied to the SiC SBDs **100**, **101**, and **102** illustrated in FIGS. **3** to **5**, and FIG. **7** shows horizontal electric fields. The vertical electric field is an electric field distributed between the Schottky junction and the lower surface of the first conductivity-type substrate **110** (indicated by Y-cut in FIGS. **3** through **5**), and the horizontal electric field is an electric field distributed between the lower surfaces of the two the second conductivity-type junction regions **140** (indicated by X-cut in FIGS. **3** through **5**). The known JBS diode to be compared has the design parameters described in FIGS. **3** through **5**.

When seeing the graph in FIG. **6**, which shows the electric fields in the vertical direction, the electric field strength tends to decrease-increase-decrease in the vertical direction from the Schottky junction. The electric field strength decreases substantially linearly in the first conductivity-type epitaxial layer beyond an area in which the electric field is concentrated by the second conductivity-type junction region, and substantially disappears in the first conductivity-type substrate. This tendency is observed in both known JBS and the SiC SBDs **100**, **101**, and **102**. However, there is a significant difference in the electric field strength at (or near) the Schottky junction. The electric field strength at the Schottky junction is about 2e+06 V/cm or more in the known JBS diode, while about 1.9e+05 V/cm in the SiC SBD **100** having the charge injection region doped with a single concentration and about 1.5e+05 V/cm in the SiC SBD **101** having the charge injection region consisting of two multi-epitaxial layers. Even in the SiC SBD **102** having the charge injection region consisting of five multi-epitaxial layers, the electric field strength at the Schottky junction decreases to about 1.3e+05 V/cm. The electric field strength at the Schottky junction is a dominant factor that causes leakage current of the Schottky diode, and the leakage current decreases as the electric field strength decreases.

The same phenomenon is also observed in FIG. **7** showing the electric fields in the horizontal direction. In the known JBS diode in which the distance between the second conductivity-type junction regions is relatively large, the electric field strength between opposite junction regions remains substantially constant due to the distance between the opposite junction regions. On the other hand, as the width of the second conductivity-type junction regions increases, the distance between the junction regions with the maximum electric field strength decreases. Due to this, the electric fields formed between the junction regions continue to decrease, so that the electric field strength at the center of the cell becomes relatively weaker than that of the known JBS diode. Therefore, the strength of the electric field applied to the Schottky junction is relatively reduced compared to the electric field in the known JBS diode.

FIG. **8** shows leakage currents, and FIG. **9** shows breakdown voltages in the SiC SBDs **100**, **101**, and **102** illustrated in FIGS. **3** through **5**. As predicted from the electric field strength graphs of FIGS. **6** and **7**, it can be seen that the increase rate of leakage currents of the SiC SBDs **100**, **101**, and **102** in which the charge injection region **130** may be formed is significantly lower than that of the known JBS diode. In addition, when seeing the breakdown voltages, the breakdown voltage of the known JBS diode is about −1,430 V, whereas the breakdown voltages of the SiC SBDs **100**, **101**, and **102** in which the charge injection region **130** may be formed is about −1,700 V.

FIGS. **10** through **13** are graphs exemplarily illustrating electrical characteristics according to a change in doping concentration of the SiC SBD shown in FIG. **5**.

The doping concentration of the SiC SBD is changed to 60%, 80%, 120% and 140% of a reference concentration profile based on the doping concentration illustrated in FIG. **5**. The depths of the first to fifth multi-epitaxial layers **130***a* though **130***e* and RJBS are not changed. The concentration profiles increased and decreased in increments of 20% are shown in Table 1. The design parameters except for the doping concentration profile are the same as those of the embodiment illustrated in FIG. **5**.

TABLE 1

	Fifth multi-epitaxial layer	Fourth multi-epitaxial layer	Third multi-epitaxial layer	Second multi-epitaxial layer	First multi-epitaxial layer	First conductivity-type epitaxial layer
Epi Depth (μm)	0.3	0.2	0.2	0.15	0.1	11
Epi Doping (cm^{-2}) [60%]	6.0E16	4.8E16	3.6E16	2.4E16	1.2E16	1E16
Epi Doping (cm^{-2}) [80%]	8.0E16	6.4E16	4.8E16	3.2E16	1.6E16	1E16
Epi Doping (cm^{-2}) [Ref]	1.0E17	8.0E16	6.0E16	4.0E16	2.0E16	1E16
Epi Doping (cm^{-2}) [120%]	1.2E17	9.6E16	7.2E16	4.8E16	2.4E16	1E16
Epi Doping (cm^{-2}) [140%]	1.4E17	1.12E17	8.4E16	5.6E17	2.8E16	1E16

FIG. **10** shows the characteristics of the forward voltage V_F and the forward current I_F of the SiC SBDs formed according to the five concentration profiles illustrated in Table 1. As illustrated, it can be seen that the change in the doping concentration does not significantly affect the forward voltage-current characteristics.

FIG. **11** shows vertical electric fields of the SiC SBDs formed according to five concentration profiles illustrated in Table 1, and FIG. **12** shows horizontal electric fields. Referring to FIG. **11**, it can be seen that the electric field strength at the Schottky junction decreases as the doping concentration increases. It can be seen that for all of the vertical electric fields of the SiC SBDs according to the five concentration profiles, the electric field strength at the Schottky junction is weaker than that case that the charge injection region may be formed at a single concentration. On the other hand, it can be seen that the electric field strength at the Schottky junction in the vertical electric fields at the relatively low concentration profiles (60% and 80%) is relatively high or almost similar than the case in which the charge injection region is formed of two multi-epitaxial layers. As previously seen in FIG. **6**, the electric field strength in the vertical direction of the SiC SBD having the charge injection region is minimal near the lower edge of the second conductivity-type junction region, then increases rapidly to maximum, and decreases almost linearly across the first conductivity-type epitaxial layer.

Referring to FIG. **12**, the horizontal electric fields of the SiC SBDs according to the five concentration profiles tend to have maximum electric field strengths of which width between them is narrow and extends vertically downward. Based on the electric field strength according to the reference concentration profile, the horizontal electric fields according to the relatively low concentration profiles have a relatively wide width between the maximum electric field strengths, whereas the horizontal electric fields according to the relatively high concentration profiles have a relatively narrow width between the maximum electric field strengths. It can be seen that as the doping concentration increases, the electric field strength applied between the junction regions increases.

Referring to FIG. **13**, as predicted from FIGS. **11** and **12**, it can be seen that the leakage current decreases as the doping concentration increases. Based on the leakage current curve according to the reference concentration profile as a reference, for relatively low concentration profiles, the increase rate of the leakage current is greater than that of the reference concentration profile and increases almost linearly over a certain range. On the other hand, in the relatively high concentration profiles, although the magnitude of the leakage current as a whole is smaller than that of the reference concentration profile, it tends to increase non-linearly.

FIG. **14** and FIG. **15** are graphs exemplarily illustrating a function of the charge injection region.

The charge injection region increases the current density. When the charge injection region may be formed over the entire upper portion of the first conductivity-type epitaxial layer, the amount of current passing through the first conductivity-type epitaxial layer increases, and a thermal runaway phenomenon may occur. Thermal runaway may increase the temperature of the package and cause damage to the surrounding area, and in particular, it is the dominant factor that increases the leakage current. Accordingly, the second conductivity-type junction region must be formed in the charge injection region to prevent excessive current from flowing.

The degree to which the charge injection region increases the current density may vary according to the concentration profile of the multi-epitaxial layer. FIG. **14** shows the result of adjusting the RJBS so that the forward voltage V_F is about 1.18 V and the forward current I_F is about 10 A. For the five concentration profiles with the depth of the second conductivity-type junction region equal to about 0.6 μm, RJBS for having the forward voltage V_F of about 1.18 V and the forward current I_F of about 10 A for each concentration profile is determined. As shown, for the concentration profiles relatively lower than the reference concentration profile, RJBS should be decreased, and conversely, for the relatively high concentration profiles, RJBS should be increased, so that VF=1.18 V and IF=10 can be satisfied. From these measurement results, it can be confirmed that the charge injection region increases the current density.

Meanwhile, FIG. **15** shows the forward voltage V_F when the forward current I_F becomes about 10 A while RJBS is fixed at 2.1 μm. For the five concentration profiles in which the depth of the second conductivity-type junction region is equal to about 0.6 μm, the forward voltages V_F when the forward current I_F is about 10 A are measured. As shown, it can be seen that as the concentration profile increases, the forward voltage V_F decreases. As with FIG. **14**, it can be confirmed that the charge injection region increases the current density from these measurement results.

FIG. **16** and FIG. **17** are graphs exemplarily illustrating electrical characteristics according to the depth of the second conductivity-type junction region.

FIG. **16** shows measurements of leakage currents for each of five concentration profiles, with the depth of the charge injection region fixed at about 0.95 μm, while RJBS is changed to 2.2, 2.4, 2.6, 2.8, 2.9 μm and the depth of the second conductive junction region is changed to about 0.4, 0.6, and 0.8 μm. Leakage currents are measured when the reverse voltage V_B is about −1,200 V. Graph **200** represents RJBS, graph **201** represents measurements of leakage current when the depth of the second conductivity-type junction region is about 0.4 μm, graph **202** represents measurements of leakage current when the depth of the second conductivity-type junction region is about 0.6 μm, and graph **203** represents measurements of the leakage current when the depth of the second conductivity-type junction region is about 0.8 μm. For example, when RJBS is 2.2 μm, the measured leakage current tends to decrease as the second conductivity-type junction region becomes deeper.

On the other hand, the leakage current tends to decrease as the doping concentration increases or the RJBS increases.

FIG. **17** shows measurements of the forward voltages V_F for each of five concentration profiles, with the depth of the charge injection region fixed at about 0.95 μm, while RJBS is changed to 2.2, 2.4, 2.6, 2.8, 2.9 μm and the depth of the second conductive junction region is changed to about 0.4, 0.6, and 0.8 μm. The forward voltages V_F are measured when the forward current I_F is about 10 A. Graph **210** represents RJBS, graph **211** represents measurements of the forward voltage V_F when the depth of the second conductivity-type junction region is about 0.4 μm, graph **212** represents measurements of the forward voltage V_F when the depth of the second conductivity-type junction region is about 0.6 μm, and graph **213** represents measurements of the forward voltage V_F when the depth of the second conductivity-type junction region is about 0.8 μm. For example, when RJBS is about 2.2 μm, the measured forward voltage V_F tends to increase as the second conductivity-type junction region becomes deeper. On the other hand, the forward voltage V_F tends to increase as the doping concentration increases or RJBS increases.

The above description of the invention is exemplary, and those skilled in the art can understand that the invention can be modified in other forms without changing the technical concept or the essential feature of the invention. Therefore, it should be understood that the above-mentioned embodiments are exemplary in all respects but are not definitive.

The scope of the invention is defined by the appended claims, not by the above detailed description, and it should be construed that all changes or modifications derived from the meanings and scope of the claims and equivalent concepts thereof are included in the scope of the invention.

What is claimed is:

1. A unit cell constituting an active area of silicon carbide junction barrier Schottky diode, comprising:

a first conductivity-type substrate;

a first conductivity-type epitaxial layer, being formed by epitaxial growth of silicon carbide doped with a first conductivity-type impurity on the first conductivity-type substrate;

a charge injection region, being defined by a second conductivity-type junction region that is formed by ion implantation of a second conductivity-type impurity into a first conductivity-type charge injection layer that is formed by epitaxial growth of silicon carbide doped at a concentration of the first conductivity-type impurity higher than that of the first conductivity-type epitaxial layer on the first conductivity-type epitaxial layer;

a Schottky metal layer, being formed on the charge injection region and the second conductivity-type junction region;

an anode electrode, being formed on the Schottky metal layer; and

a cathode electrode, being formed under the first conductivity-type substrate;

wherein the first conductivity-type charge injection layer comprises two or more multi-epitaxial layers having different first conductivity-type impurity concentrations;

wherein the first conductivity-type impurity concentrations of the two or more multi-epitaxial layers according to a reference concentration profile decrease in a direction toward the first conductivity-type epitaxial layer.

2. The unit cell according to claim **1**, wherein an area occupied by the second conductivity-type junction region in the unit cell is in a range between 62% and 82%.

3. The unit cell according to claim **1**, wherein the reference concentration profile defines the first conductivity-type impurity concentrations of the two or more multi-epitaxial layers.

4. The unit cell according to claim **3**, wherein the reference concentration profile decreases the impurity concentration by 20% in the direction toward the first conductivity-type epitaxial layer.

5. The unit cell according to claim **3**, wherein the first conductivity-type impurity concentrations of each of the two or more multi-epitaxial layers are equally increased by a maximum of 40% or a decrease of a minimum of −40% in the reference concentration profile.

6. The unit cell according to claim **1**, wherein the depths of two or more multi-epitaxial layers decrease in the direction toward the first conductivity-type epitaxial layer.

7. The unit cell according to claim **1**, wherein a depth of the second conductivity-type junction region is in a range between 60% and 100% of a depth of the charge injection region.

8. A silicon carbide junction barrier Schottky diode having an active area, comprising:

a first conductivity-type substrate;

a first conductivity-type epitaxial layer, being formed by epitaxial growth of silicon carbide doped with a first conductivity-type impurity on the first conductivity-type substrate;

a charge injection region, being formed on the first conductivity-type epitaxial layer and doped at a concentration of the first conductivity-type impurity higher than that of the first conductivity-type epitaxial layer;

a second conductivity-type junction region, being formed on the first conductivity-type epitaxial layer so as to contact the charge injection region;

a Schottky metal layer, being formed on the charge injection region and the second conductivity-type junction region;

an anode electrode, being formed on the Schottky metal layer; and

a cathode electrode, being formed under the first conductivity-type substrate;

wherein the first conductivity-type charge injection layer comprises two or more multi-epitaxial layers having different first conductivity-type impurity concentrations;

wherein the first conductivity-type impurity concentrations of the two or more multi-epitaxial layers according to a reference concentration profile decrease in a direction toward the first conductivity-type epitaxial layer.

9. The silicon carbide junction barrier Schottky diode according to claim **8**, wherein an area occupied by the second conductivity-type junction region in the active area is in a range between 62% and 82%.

10. The silicon carbide junction barrier Schottky diode according to claim **8**, wherein the reference concentration profile defines the first conductivity-type impurity concentrations of the two or more multi-epitaxial layers.

11. The silicon carbide junction barrier Schottky diode according to claim **8**, wherein the depths of two or more multi-epitaxial layers decrease in the direction toward the first conductivity-type epitaxial layer.

12. The silicon carbide junction barrier Schottky diode according to claim **8**, wherein a depth of the second conductivity-type junction region is in a range between 60% and 100% of a depth of the charge injection region.

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